

MESSIER 12 GLOBULAR CLUSTER

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1 Introduction to Hubble and Messier 12

The Hubble Space Telescope (HST) is a 2.4-meter space-based observatory launched by NASA in 1990 into low-Earth orbit. Operating above Earth's atmosphere, HST is capable of capturing high-resolution images in ultraviolet, visible, and near-infrared wavelengths without atmospheric distortion. Over the past three decades, Hubble has played a major role in modern astronomy by contributing to the study of galaxies, nebulae, black holes, exoplanets, and globular clusters.

Messier 12 (M12) is a globular cluster located in the constellation Ophiuchus at a distance of approximately 23,000 light-years from Earth. Discovered by Charles Messier in 1764, M12 contains a dense population of old stars gravitationally bound within a nearly spherical stellar system. Globular clusters such as M12 are important astrophysical laboratories for studying stellar evolution, stellar dynamics, and the structure of the Milky Way galaxy.

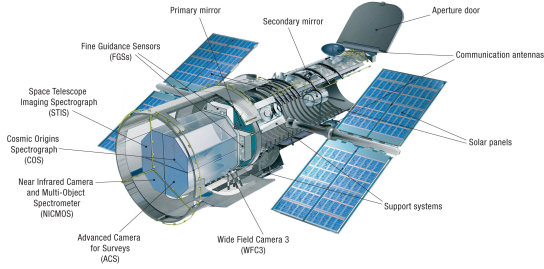
Astronomical studies suggest that M12 contains fewer low-mass stars than expected, possibly due to gravitational stripping caused by repeated interactions with dense regions of the Milky Way during its galactic orbit. The cluster also contains binary star systems and several X-ray sources produced through stellar interactions within the dense stellar environment.

Although Hubble observations are commonly associated with globular cluster studies, the observational FITS dataset analyzed in this project was obtained using the FORS1 instrument mounted on the ESO Very Large Telescope (VLT-U1). The image was captured using the V-Bessell optical filter and stored in FITS (Flexible Image Transport System) format, which is the standard data format used in observational astronomy.

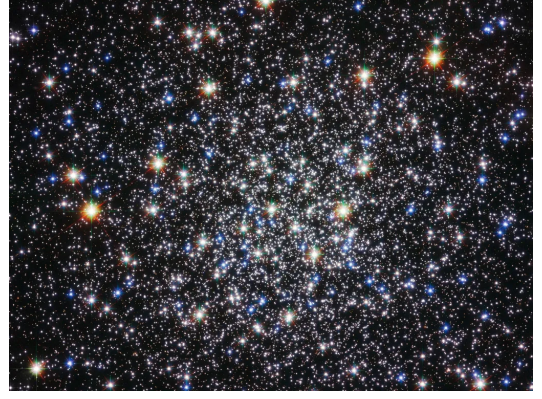
The primary objective of this project is to perform astronomical image analysis on the Messier 12 FITS image using Python-based scientific libraries. The workflow includes image visualization, background estimation using Sigma Clipping, stellar source detection using [DAOSStarFinder](#), and photometric flux analysis of the detected stellar population.

2 Procedure

The complete workflow followed in this project is shown in Figure 2. The analysis pipeline consists of FITS image loading, visualization, background estimation, stellar source detection, and flux analysis.



(a) Hubble Space Telescope



(b) Globular Cluster M12

Figure 1: *Left: The Hubble Space Telescope (Image source: Mikulski Archive for Space Telescopes). Right: The globular cluster M12 captured by the Hubble Space Telescope (Image source: NASA).*

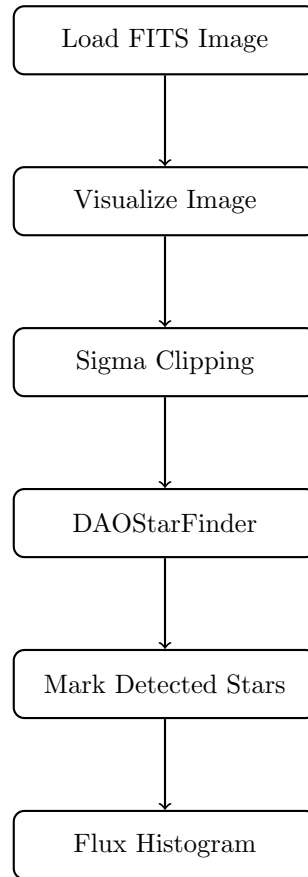


Figure 2: Workflow of the Messier 12 analysis pipeline.

2.1 Loading the FITS File

The astronomical image stored in the `vcomb.fits` file was loaded using the `astropy.io.fits` module. The science image was extracted from the primary HDU as a 2048×2048 floating-point array for further analysis.

2.2 Image Visualization

The FITS image was visualized using `matplotlib.pyplot.imshow()` with the `inferno` colormap. Since astronomical images contain a large dynamic range, logarithmic normalization using `LogNorm()` was applied to enhance faint stellar structures while preserving bright regions.

2.3 Background Estimation using Sigma Clipping

The background statistics of the image were estimated using the Astropy function `sigma_clipped_stats()`. Sigma clipping iteratively removes outlier pixels beyond a specified threshold, thereby reducing the influence of bright stars during background estimation.

A clipping threshold of $\sigma = 3.0$ was used to compute the background mean, median, and standard deviation values.

2.4 Star Detection using DAOSTarFinder

Stellar sources were detected using the `DAOSTarFinder` algorithm from the Photutils package. The detector was initialized with a Full Width at Half Maximum (FWHM) value of 3.0 pixels and a detection threshold equal to five times the estimated background standard deviation.

Before source extraction, the median background value was subtracted from the image to suppress background contributions and improve detection accuracy.

2.5 Marking Detected Stars

The centroid positions returned by `DAOSTarFinder` were used to overlay circular apertures on the logarithmically scaled image using `CircularAperture`. The detected stars were marked with green circular boundaries for visualization of stellar positions across the cluster field.

2.6 Flux Analysis

The flux values of the detected stars were extracted from the source catalog generated by `DAOSTarFinder`. A logarithmic flux histogram using $\log_{10}(\text{Flux})$ was then plotted to study the brightness distribution of stars in Messier 12. The histogram revealed that the majority of stars possess relatively low flux values, while only a small fraction of stars exhibit extremely high brightness.

3 Results and Discussion

The FITS image of Messier 12 was successfully processed and analyzed using Python-based astronomical data analysis tools. The complete analysis pipeline included image visualization, background estimation using Sigma Clipping, stellar source detection using `DAOSTarFinder`, and flux analysis.

Important observational metadata extracted from the FITS header are summarized in Table 1. The analyzed image was obtained using the FORS1 instrument mounted on the ESO Very Large Telescope (VLT) with the V-Bessell optical filter. The dataset consists of a 2048×2048 pixel image with a pixel scale of 0.2 arcsec/pixel and an average atmospheric seeing of approximately 0.64 arcsec.

Table 1: Important metadata extracted from the FITS header.

Parameter	Value
Telescope	ESO VLT (FORs1)
Filter	V-Bessell
Observation Date	1999-06-18
Exposure Time	2.0 s
Pixel Scale	0.2 arcsec/pixel
Seeing (FWHM)	0.64 arcsec
Image Size	2048 $\times$ 2048 pixels

Figure 3 shows the original FITS image displayed with logarithmic normalization. The use of `LogNorm()` improved the visibility of faint stellar structures across the globular cluster field.

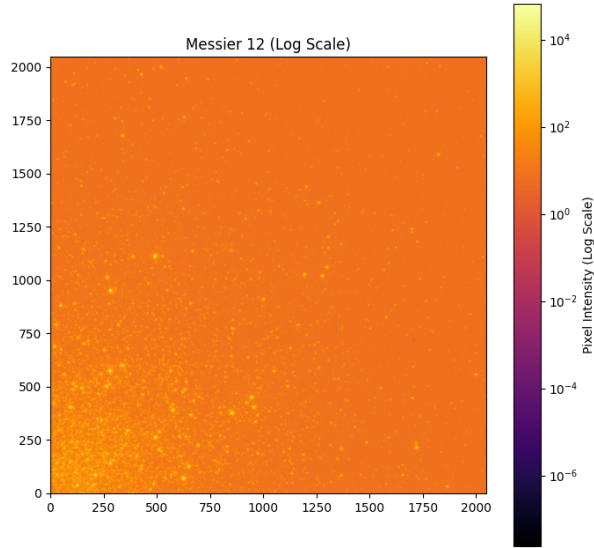


Figure 3: Log-scaled visualization of the Messier 12 FITS image.

Background statistics were estimated using `sigma_clipped_stats()`, which provided stable mean, median, and standard deviation values by reducing the influence of bright stellar outliers during iterative sigma clipping. The estimated background mean, median, and standard deviation were found to be 9.28, 9.00, and 2.26 ADU, respectively. These values were subsequently used to define a reliable detection threshold for stellar source extraction.

Figure 4 shows the stellar sources detected using `DAOSTarFinder` and marked with circular apertures. The source extraction process successfully identified stellar objects across the dense globular cluster region.

A total of **4950** stellar sources were detected in the analyzed FITS image. Table 2 summarizes the properties of the brightest and faintest detected stars.

Table 2: Properties of the brightest and faintest detected stars in Messier 12.

Parameter	Brightest Star	Faintest Star
Source ID	2669	4664
X Centroid (px)	279.25	1067.68
Y Centroid (px)	572.54	1561.23
Peak Intensity (ADU)	66207.25	11.73
Flux (ADU)	940086.25	130.45
Magnitude	-14.93	-5.29

The detected stellar population spans a very large brightness range, from faint low-flux stars to highly luminous stellar sources. This wide dynamic range is characteristic of dense globular cluster systems and motivates the use of logarithmic scaling during image visualization and flux analysis.

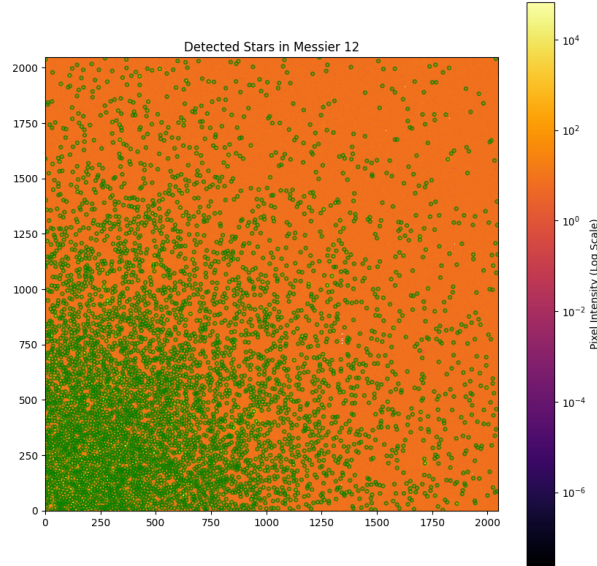


Figure 4: Detected stars in Messier 12 marked using circular apertures.

The flux values of the detected stars were analyzed using a histogram. Figure 5 shows the logarithmic flux distribution of the detected stellar population. The distribution indicates that most stars are faint, while only a small fraction exhibit very high flux values, which is typical of globular cluster systems.

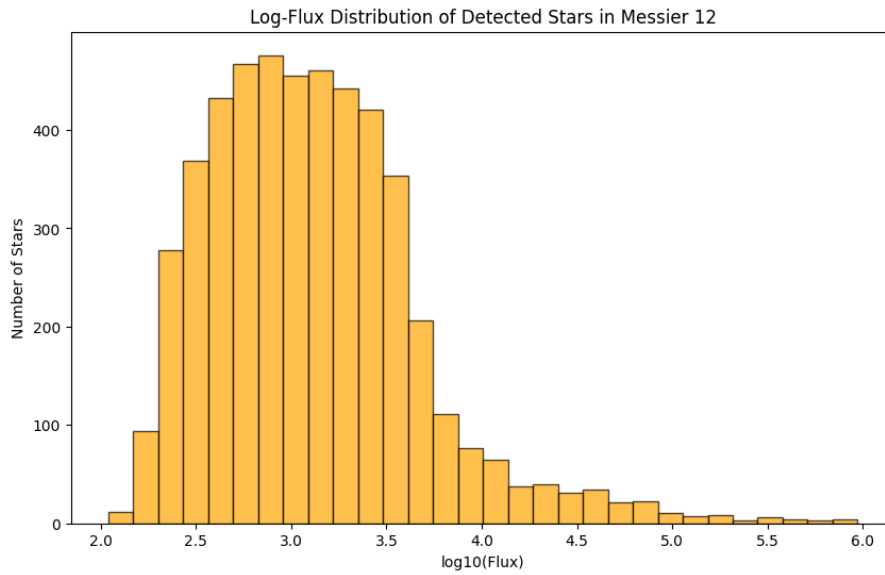


Figure 5: Logarithmic flux distribution of detected stars in Messier 12.

Overall, the project demonstrates a complete astronomical image-analysis workflow involving FITS data handling, background estimation, stellar source detection, and photometric analysis.

4 Python Code

4.1 Data Loading and Visualization

```
#importing the required libraries
import numpy as np
import matplotlib.pyplot as plt
from astropy.io import fits
from astropy.stats import sigma_clipped_stats

# Loading the FITS file
file_path = "../data/vcomb.fits"
hdul = fits.open(file_path)
header = hdul[0].header
data = hdul[0].data
hdul.close()

# Printing basic information about the data
print("Data shape:", data.shape)
print(hdul.info())
# the brightest pixel value in the data
max_pixel_value = np.max(data)
print("Brightest Pixel Value:", max_pixel_value)

# the darkest pixel value in the data
min_pixel_value = np.min(data)
print("Darkest Pixel Value:", min_pixel_value)

# the mean pixel value in the data
mean_pixel_value = np.mean(data)
print("Mean Pixel Value:", mean_pixel_value)

# Visualizing the data
```

```

plt.figure(figsize=(8,8))
plt.imshow(data, cmap='inferno', origin='lower')
plt.colorbar(label="Pixel Intensity")
plt.title("Messier 12")
plt.show()

# using Logarithmic scaling to enhance the visibility
from matplotlib.colors import LogNorm
plt.figure(figsize=(8,8))
plt.imshow(data, cmap='inferno', origin='lower', norm=LogNorm())
plt.colorbar(label="Pixel Intensity (Log Scale)")
plt.title("Messier 12 (Log Scale)")
plt.show()

```

4.2 Background Estimation and Star Detection

```

# Estimating background statistics using sigma clipping
bg_mean, bg_median, bg_std = sigma_clipped_stats(data, sigma=3.0)
print("Background Mean:", bg_mean)
print("Background Median:", bg_median)
print("Background Standard Deviation:", bg_std)

# Detecting stars using the DAOStarFinder Algorithm
from photutils.detection import DAOStarFinder

daofind = DAOStarFinder(fwhm=3.0, threshold=5.0*bg_std)
stars = daofind(data - bg_median)
for col in stars.colnames:
    if col not in ('id', 'npix'):
        stars[col].info.format = '%.2f'
stars.pprint(max_width=76)

from photutils.aperture import CircularAperture

positions = np.transpose((stars['x_centroid'], stars['y_centroid']))
apertures = CircularAperture(positions, r=7)

# Visualize the detected stars
plt.figure(figsize=(10,10))
plt.imshow(data, cmap='inferno', origin='lower', norm=LogNorm(),
           interpolation='nearest')
apertures.plot(color='green', lw=1.5, alpha=0.7)
plt.colorbar(label="Pixel Intensity (Log Scale)")
plt.title("Detected Stars in Messier 12")
plt.show()

# Finding the brightest star among the detected stars
brightest_star = stars[np.argmax(stars['peak'])]
print("Brightest Star:")
print("ID:", brightest_star['id'])
print("X Centroid:", brightest_star['x_centroid'], "px")
print("Y Centroid:", brightest_star['y_centroid'], "px")
print("Peak Intensity:", brightest_star['peak'], "ADU")
print("flux:", brightest_star['flux'], "ADU")
print("magnitude:", brightest_star['mag'])

```

```

# Finding the faintest star among the detected stars
faintest_star = stars[np.argmin(stars['peak'])]
print("Faintest Star:")
print("ID:", faintest_star['id'])
print("X Centroid:", faintest_star['x_centroid'], "px")
print("Y Centroid:", faintest_star['y_centroid'], "px")
print("Peak Intensity:", faintest_star['peak'], "ADU")
print("flux:", faintest_star['flux'], "ADU")
print("magnitude:", faintest_star['mag'])

```

4.3 Histogram of the flux of the detected stars

```

# Extracting the flux values of the detected sources and plotting a
  histogram to visualize the distribution of fluxes among the detected
  stars
fluxes = stars['flux']

# Convert flux values to logarithmic scale
log_fluxes = np.log10(fluxes)

# Plot histogram
plt.figure(figsize=(10,6))
plt.hist(log_fluxes, bins=30, color='orange', edgecolor='black', alpha=0.7)
plt.xlabel('log10(Flux)')
plt.ylabel('Number of Stars')
plt.title('Log-Flux Distribution of Detected Stars in Messier 12')
plt.show()

```


5 References

1. NASA Science Mission Directorate, *Messier 12 – Hubble Messier Catalog*. Available at: <https://science.nasa.gov/mission/hubble/science/explore-the-night-sky/hubble-messier-catalog/messier-12/>
2. Isabell Hudnall, *Photutils.detection — Detecting Stars with DAOStarFinder*, YouTube video tutorial. Available at: https://www.youtube.com/watch?v=vxMoG2K_n5o
3. Hubble Space Telescope (HST) Archive, *NASA/ESA Hubble Space Telescope Data Archive*. Available at: <https://archive.stsci.edu/hst/>

Source Code

The complete Python implementation, Jupyter notebook, and generated analysis files for this project are available in the following GitHub repository: https://github.com/arsanjeevk/M12_Globular_Cluster